Skill-21

```
himasree@LAPTOP-41COT5AA:~/OSLab$ nano integration_test.c
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc -Wall -Wextra integration_test.c -o integration_test
himasree@LAPTOP-41COT5AA:~/OSLab$ ./integration_test
Integration testing started.
Parent waiting for child process.
Child process executing command.
total 2092
-rwxr-xr-x 1 himasree himasree 16264 Aug 27 09:37 Apipes
-rw-r--r-- 1 himasree himasree   722 Aug 28 04:51 Apipes.c
-rw-r--r-- 1 himasree himasree   218 Aug 28 04:54 Apipes2.c
-rwxr-xr-x 1 himasree himasree 16048 Aug 28 04:53 Appipes2
-rwxr-xr-x 1 himasree himasree 16056 Aug 28 05:00 FIFO_CREATE
-rw-r--r-- 1 himasree himasree   220 Aug 28 04:59 FIFO_CREATE.c
-rwxr-xr-x 1 himasree himasree 16144 Aug 28 05:01 FIFO_READ
-rw-r--r-- 1 himasree himasree   287 Aug 28 05:01 FIFO_READ.c
-rwxr-xr-x 1 himasree himasree 16184 Aug 28 05:04 FIFO_WRITE
-rw-r--r-- 1 himasree himasree   292 Aug 28 05:02 FIFO_WRITE.c
drwxr-xr-x 2 himasree himasree  4096 Aug  6 08:36 LinuxLab
drwxr-xr-x 3 himasree himasree  4096 Aug  1 03:05 MYProject
drwxr-xr-x 7 himasree himasree  4096 Aug  1 03:33 MyProject
drwxr-xr-x 2 himasree himasree  4096 Aug 27 09:27 algorithms
```

```
Integration test successful.
himasree@LAPTOP-41COT5AA:~/OSLab$ nano error_handling.c
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc -Wall -Wextra error_handling.c -o error_handling
himasree@LAPTOP-41COT5AA:~/OSLab$ ./error_handling
MiniShell> MiniShell> ls
MiniShell> invalidcommand
MiniShell> pwd
MiniShell> exitInvalid command or execution failed: No such file or directory
Error handled. Enter another command.
MiniShell> Invalid command or execution failed: No such file or directory
Error handled. Enter another command.
MiniShell> Invalid command or execution failed: No such file or directory
Error handled. Enter another command.
MiniShell> ls
Invalid command or execution failed: No such file or directory
Error handled. Enter another command.
MiniShell>
```

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char command[100];
    pid_t pid;
    int status;

    while (1)
    {
        printf("MiniShell> ");
        fflush(stdout);

        if (fgets(command, sizeof(command), stdin) == NULL)
        {
            printf("\nInput ended. Exiting.\n");
            break;
        }

        command[strcspn(command, "\n")] = '\0';
```

```
        if (pid == 0)
        {
            execlp(command, command, (char *)NULL);

            perror("Invalid command or execution failed");
            exit(127);
        }
        else
        {
            if (waitpid(pid, &status, 0) == -1)
            {
                perror("Waiting for child failed");
                continue;
            }

            if (WIFEXITED(status) && WEXITSTATUS(status) == 127)
                printf("Error handled. Enter another command.\n");
            else
                printf("Command execution completed.\n");
        }
    }

    printf("MiniShell terminated safely.\n");
    return 0;
}
```

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;
    int status;

    printf("Integration testing started.\n");

    pid = fork();

    if (pid < 0)
    {
        perror("Process creation failed");
        return 1;
    }

    if (pid == 0)
    {
        printf("Child process executing command.\n");
        execlp("ls", "ls", "-l", (char *)NULL);
```

```
    {
        printf("Child process executing command.\n");
        execlp("ls", "ls", "-l", (char *)NULL);

        perror("Command execution failed");
        exit(1);
    }
    else
    {
        printf("Parent waiting for child process.\n");

        if (waitpid(pid, &status, 0) == -1)
        {
            perror("waitpid failed");
            return 1;
        }

        if (WIFEXITED(status) && WEXITSTATUS(status) == 0)
            printf("Integration test successful.\n");
        else
            printf("Integration test failed.\n");
    }

    return 0;
}
```